



TRINITY GRAMMAR SCHOOL

Mathematics Department

2012

HALF YEARLY EXAMINATION

ASSESSMENT TASK 3

Year 11

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – $2\frac{1}{2}$ hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used.
- Show all necessary working in Questions 11 – 16
- Write your NAME **and** Class Teacher on the writing booklet(s) **or** sheet(s) submitted.
- Weighting 30%

Total marks – 100

Section I

Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 6 – 12

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 15 minutes for this section

Section I 10 marks

- Shade the correct response on the answer sheet provided
 - Each question is worth 1 mark
-

1. Simplify: $5^{m+1} \times 5^{m+3}$

- a) 25^{2m+4}
- b) 5^{m^2+3}
- c) 5^{2m+4}
- d) 5^{m^2+4m+3}

2. Solve for a : $\frac{a}{10} + \frac{a}{5} = 3$

- a) $a = 10$
- b) $a = 1$
- c) $a = 15$
- d) $a = \frac{1}{2}$

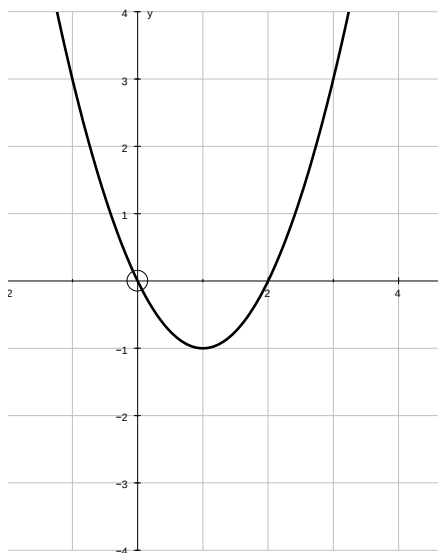
3. Solve for n : $\frac{n-7}{4} - \frac{n-1}{9} = 1$

- a) $n = 12$
- b) $n = 19$
- c) $n = 12\frac{3}{5}$
- d) $n = 13\frac{3}{5}$

4. After a 20% discount on a fridge, Rudy paid \$429 for his new fridge. Calculate the original price of the fridge before the discount correct to the nearest dollar.

- a) \$2145
- b) \$536
- c) \$515
- d) \$343

5. The equation of the following function is:



- a) $y = x^2 + x - 1$
- b) $y = x^2 + 2x$
- c) $y = x^2 + 2x - 2$
- d) $y = x^2 - 2x$

6. Factorise: $y^3 + 8z^3$

a) $(y + 2z)(y^2 - 2yz + 4z^2)$

b) $y + 2z \quad y^2 + 2yz + 4z^2$

c) $y + 2z \quad y^2 + 4yz + 4z^2$

d) $(y - 2z)(y^2 + 2yz + 4z^2)$

7. What is the equation of a line passing through (2,-1) and parallel to the line represented by the equation $y = 2x + 1$

a) $y = -\frac{1}{2}x$

b) $y = -\frac{1}{2}x + 1$

c) $y = 2x - 5$

d) $y = 2x - 1$

8.

a) 35°

b) 41°

c) 44°

d) 46°

9.

The length of side AC is closest to:

- a) 57 m
- b) 67 m
- c) 77 m
- d) 81 m

10.

The length of the longer diagonal of this parallelogram is closest to:

- a) 26.5 cm
- b) 30.1 cm
- c) 30.8 cm
- d) 34.5 cm

End of Section I

Section II 90 marks

- Begin each question in a **new** writing booklet or on a new answer sheet
 - Show all necessary working
 - Each question is worth 15 marks
-

Question 11 (15 marks)

- (a) Evaluate $\frac{3 \cdot 4^2 + 2 \cdot 5^4}{5 \cdot 56 \times 10^{-3}}$ correct to 3 significant figures. 1
- (b) Simplify $36x^4y^{-2} \div (6xy)^3$ 2
- (c) Expand and simplify $8(a - 2b) - 6(2a - 3b)$ 2
- (d) Express $0.\dot{7}\dot{3}$ as a fraction in its simplest form. 2
- (e) Simplify $\frac{a^2 - 2a - 3}{a^3 + 1}$ 2
- (f) Simplify $4\sqrt{18} + 3\sqrt{12} - 2\sqrt{50}$ 2
- (g) Given the equation $v^2 = u^2 + 2as$
- (i) make u the subject of the equation. 1
- (ii) Hence or otherwise find the value(s) of u when,
 $v = 12, a = -14, s = 4$ 1
- (h) Factorise fully $x^3 - y^3 + y^2 - x^2$ 2

Question 12 (15 marks)**Begin a NEW answer booklet or answer sheet**

(a) Solve $1 \leq \frac{2x-3}{5} < 5$, and plot the solution on a number line. 3

(b) If $x - \sqrt{y} = (3 - \sqrt{2})^2$, find the value of x and y . 3

(c) Fully factorise $4x^3 - 32$ 2

(d) Simplify the following

i) $\frac{x^2 - 2x}{y^2 - 4y + 4} \times \frac{y-2}{x}$ 2

ii) $\frac{3}{x-y} + \frac{x}{x^2 - y^2}$ 3

(e) Simplify $\frac{\sqrt{6}}{\sqrt{5} + \sqrt{2}}$ by rationalising the denominator. 2

Question 13 (15 marks)**Begin a NEW answer booklet or answer sheet**

(a) Solve simultaneously

$$x - y = 2$$

$$y = x^2 - 4$$

3

(b) Solve the following equations

(i) $6x^2 + 13x - 5 = 0$

2

(ii) $\frac{x}{6} - \frac{x-1}{3} = 5$

3

(iii) $|1 - 3x| = 8$

2

(c) Solve the following inequation

$$\left| \frac{x}{2} - 1 \right| < 5$$

3(d) By using the quadratic formula, or otherwise solve the following equation,
leaving answer in simplified surd form.

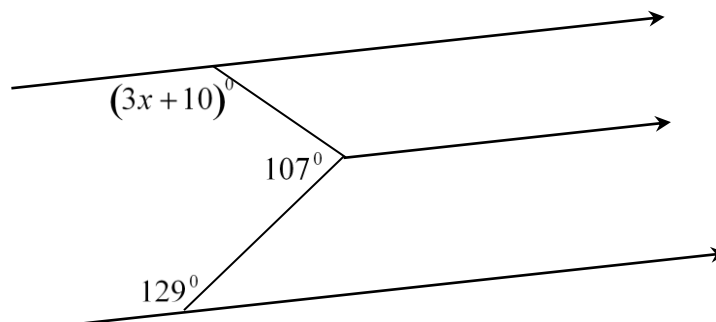
$$-9x + 3x^2 = 2$$

2

Question 14 (15 marks)

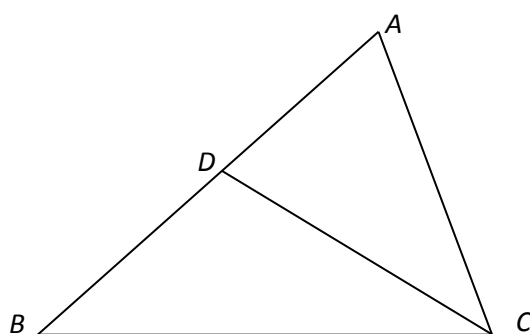
Begin a NEW answer booklet or answer sheet

- (a) Redraw the diagram below and then find the value of x , giving all reasons.



2

- (b)



- (i) Given $\angle ABC = \angle ACD$, show that $\triangle ABC \sim \triangle ACD$. 2
- (ii) Hence show that $AC^2 = AB \times AD$. 1
- (iii) Find the length of AB if $AC=6\text{cm}$, $AD=4\text{cm}$. 1
- (c) Find the largest possible domain for the function $y = \sqrt{2x-3}$. 2

(d) For the function $f(x) = \begin{cases} -1 & x \leq -2 \\ 2x & -2 < x < 0 \\ 3 - x^2 & x \geq 0 \end{cases}$,

evaluate $f(-2) + f\left(\frac{1}{2}\right) + 2f(4)$

3

(e) Sketch the following graphs showing all the essential features.

(i) $y = x^2 - 4x$

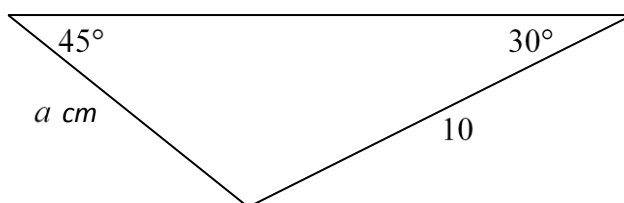
2

(ii) $y = |x - 1|$

2

Question 15 (15 marks)**Begin a NEW answer booklet or answer sheet**

- (a) Find the **exact** value of
- (i) $\cos 135^\circ$ 1
- (ii) $\tan(-60^\circ)$ 1
- (b) If $\tan \theta = -\frac{3}{4}$ and $\sin \theta < 0$, find $\cos \theta$ 2
- (c) Solve $2\sin \theta + 1 = 0$ for $0 \leq \theta \leq 360^\circ$ 2
- (d) Simplify: $\frac{\sin(90^\circ - \theta)}{\sin(180^\circ - \theta)}$ 2
- (e) A ship sets sail from Sydney for 200 km on a bearing of 040° .
It then turns on to a bearing of 157° and sails for another 345 km.
- i) Draw a neat sketch of the information. 1
- ii) How far is the ship from Sydney, to the nearest km? 2
- iii) What is the bearing of the ship from Sydney, to the nearest degree? 2
- (f) Find the **exact** value of a in the diagram below. 2



Question 16 (15 marks)**Begin a NEW answer booklet or answer sheet**

- (a) (i) Show whether the function $f(x) = \frac{1}{x-1} + 2$ is odd, even or neither. 3
- (ii) Neatly and clearly sketch the graph of $y = \frac{1}{x-1} + 2$,
showing all intercepts and asymptotes. 3
- (iii) State the domain and range of the function, $f(x) = \frac{1}{x-1} + 2$. 2
- (iv) Write down the equation of the vertical asymptote of $f(x) = \frac{1}{x-1} + 2$. 1
- (b) Show that $\tan \theta - \frac{\sin^3 \theta}{\cos \theta} = \sin \theta \cos \theta$ 2
- (c) (i) Show that the equation for the area of an equilateral triangle,
with side length a , can be written as $Area = \frac{a^2 \sqrt{3}}{4}$ 2
- (ii) If the area of the same equilateral triangle is $16\sqrt{3} \text{ cm}^2$, find the length of its
sides. 2

End of Section II

End of Examination



TRINITY GRAMMAR SCHOOL

2012, Year 11 Mathematics

Assessment Task 3

SECTION I

ANSWER SHEET

Name:

Class Teacher:

Be sure to write your answers for Section I on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. Mark only one answer for each question.

- | | | | | |
|-------------|---|---|---|---|
| Q1. | A | B | C | D |
| Q2. | A | B | C | D |
| Q3. | A | B | C | D |
| Q4. | A | B | C | D |
| Q5. | A | B | C | D |
| Q6. | A | B | C | D |
| Q7. | A | B | C | D |
| Q8. | A | B | C | D |
| Q9. | A | B | C | D |
| Q10. | A | B | C | D |



TRINITY GRAMMAR SCHOOL
2012, Year 11 Mathematics
Assessment Task 3
SECTION I
ANSWER SHEET

Name: SOLUTIONS.....

Class Teacher:

Be sure to write your answers for Section I on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. Mark only one answer for each question.

- | | | | | |
|------|----------|----------|----------|----------|
| Q1. | A | B | <u>C</u> | D |
| Q2. | <u>A</u> | B | C | D |
| Q3. | A | <u>B</u> | C | D |
| Q4. | A | <u>B</u> | C | D |
| Q5. | A | B | C | <u>D</u> |
| Q6. | <u>A</u> | B | C | D |
| Q7. | A | B | <u>C</u> | D |
| Q8. | <u>A</u> | B | C | D |
| Q9. | A | B | <u>C</u> | D |
| Q10. | A | B | C | <u>D</u> |

Question 11:

$$(a) \frac{3.4^2 + 2.5^4}{5.56 \times 10^{-3}}$$

$$= 9104.766$$

$$= 9100 \text{ (3 sf)} \checkmark$$

$$(b) \frac{36x^4y^{-2}}{(6xy)^3}$$

$$= \frac{6^2 x^4 y^{-2}}{6^3 x^3 y^3}$$

$$= \frac{xy^{-5}}{6}$$

$$= \frac{x}{6y^5} \checkmark \checkmark$$

$$(c) 8(a-2b) - 6(2a-3b)$$

$$= 8a - 16b - 12a + 18b \checkmark$$

$$= -6a + 2b \checkmark$$

$$(d) 0.7\dot{3}$$

$$x = 0.7333$$

$$10x = 7.33$$

$$100x = 73.33$$

$$\therefore 90x = 66$$

$$x = \frac{66}{90}$$

$$= \frac{11}{15} \checkmark$$

$$(e) \frac{a^2 - 2a - 3}{a^3 + 1}$$

$$= \frac{(a-3)(a+1)}{(a+1)(a^2 - a + 1)} \checkmark$$

$$= \frac{a-3}{a^2 - a + 1} \checkmark$$

$$(f) 4\sqrt{18} + 3\sqrt{12} - 2\sqrt{50}$$

$$= 12\sqrt{2} + 6\sqrt{3} - 10\sqrt{2} \checkmark$$

$$= 2\sqrt{2} + 6\sqrt{3} \checkmark$$

$$(g) \begin{aligned} V^2 &= U^2 + 2as \quad \text{ii)} \quad U = \pm \sqrt{(12)^2 - 14 \times 2 \times 4} \\ U^2 &= V^2 - 2as \\ U &= \pm \sqrt{V^2 - 2as} \end{aligned}$$

$$= \pm \sqrt{256}$$

$$= \pm 16 \checkmark$$

$$h) x^3 - y^3 - x^2 + y^2$$

$$= (x-y)(x^2 + xy + y^2) + (x^2 - y^2) \checkmark$$

$$= (x-y)(x^2 + xy + y^2) - (x-y)(x+y)$$

$$= (x-y)(x^2 + xy + y^2 - x - y) \checkmark$$

TOTAL = 15

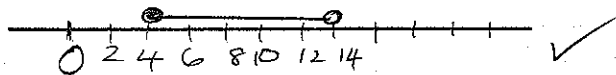
Question 12:

a) $1 \leq \frac{2x-3}{5} < 5$

$$5 \leq 2x-3 < 25 \quad \checkmark$$

$$8 \leq 2x < 28$$

$$4 \leq x < 14 \quad \checkmark$$



b) $x - \sqrt{y} = (3 - \sqrt{2})^2$

$$= 9 - 6\sqrt{2} + 2 \quad \checkmark$$

$$= 11 - 6\sqrt{2}$$

$$= 11 - \sqrt{72} \quad \checkmark$$

$$x = 11 \quad y = 72 \quad \checkmark$$

c) $4x^3 - 32$

$$= 4(x^3 - 8) \quad \checkmark$$

$$= 4(x-2)(x^2 + 2x + 4) \quad \checkmark$$

d) i) $\frac{x^2 - 2x}{y^2 - 4y + 4} \times \frac{y-2}{x}$

$$= \frac{x(x-2)}{(y-2)(y-2)} \times \frac{(y-2)}{x} \quad \checkmark$$

$$= \frac{x-2}{y-2} \quad \checkmark$$

ii) $\frac{3}{x-y} + \frac{x}{x^2 - y^2}$

$$= \frac{3}{x-y} + \frac{x}{(x+y)(x-y)} \quad \checkmark$$

$$= \frac{3(x+y) + x}{(x+y)(x-y)} \quad \checkmark$$

$$= \frac{3x + 3y + x}{(x+y)(x-y)}$$

$$= \frac{4x + 3y}{(x+y)(x-y)} \quad \checkmark$$

e) $\frac{\sqrt{6}}{\sqrt{5} + \sqrt{2}}$

$$= \frac{\sqrt{6}}{\sqrt{5} + \sqrt{2}} \times \frac{(\sqrt{5} - \sqrt{2})}{(\sqrt{5} - \sqrt{2})} \quad \checkmark$$

$$= \frac{\sqrt{30} - 2\sqrt{6}}{5 - 2}$$

$$= \frac{\sqrt{30} - 2\sqrt{6}}{3} \quad \checkmark$$

TOTAL: 15

Question 13:

a) $x - y = 2$ (1)

$y = x^2 - 4$ (2)

from (1) $y = x - 2$

∴ subst into (2)

$x - 2 = x^2 - 4$ ✓

$0 = x^2 - x - 2$ ✓

$0 = (x - 2)(x + 1)$ ✓

$\frac{x-2}{y=0}$ OR $\frac{x+1}{y=-3}$ ✓

b) i) $6x^2 + 13x - 5 = 0$ (3)

$6x^2 + 13x - 5 = (2x + 5)(3x - 1) = 0$ ✓

$x = -\frac{5}{2}$ OR $x = \frac{1}{3}$ (2)

ii) $\frac{x}{6} - \frac{x-1}{3} = 5$ (L.O.D = 6) ✓

$x - 2(x-1) = 30$ ✓

$x - 2x + 2 = 30$

$-x + 2 = 30$

$-x = 28$

$x = -28$ ✓

iii) $|1 - 3x| = 8$ (3)

$1 - 3x = 8$ OR $-(1 - 3x) = 8$

$-3x = 7$

$x = -\frac{7}{3}$ ✓

$-1 + 3x = 8$

$3x = 9$

$x = 3$ ✓

(2)

c) $\left| \frac{x}{2} - 1 \right| < 5$

$-5 < \frac{x}{2} - 1 < 5$ ✓

$-4 < \frac{x}{2} < 6$ ✓

$-8 < x < 12$ ✓

(3)

d) $-9x + 3x^2 = 2$

$3x^2 - 9x - 2 = 0$

$x = \frac{-(-9) \pm \sqrt{(-9)^2 - 4(3)(-2)}}{2 \times 3}$ ✓

$= \frac{9 \pm \sqrt{81 + 24}}{6}$

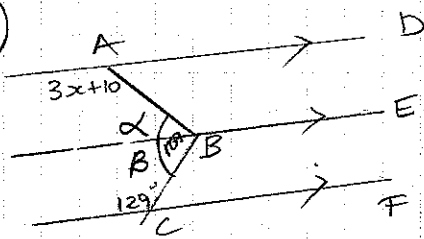
$= \frac{9 \pm \sqrt{105}}{6}$ ✓

(2)

TOTAL = 15

Question 14:

a)



$$\alpha + \beta = 107^\circ$$

$$\beta + 129^\circ = 180^\circ \quad (\text{co-int. } \angle\text{'s in } \parallel \text{ lines})$$

$$\beta = 51^\circ$$

$$107^\circ - 51^\circ = 56^\circ \quad (\text{adj. } \angle\text{'s})$$

$$\therefore 3x + 10^\circ + 56^\circ = 180^\circ \quad (\text{co-int. } \angle\text{'s in } \parallel \text{ lines})$$

$$3x = 114^\circ$$

$$x = 38^\circ$$

OR

$$\angle ABE = 3x + 10^\circ \quad (\text{alt } \angle\text{'s in } \parallel \text{ lines})$$

$$\angle CBE = 129^\circ \quad (\text{alt } \angle\text{'s in } \parallel \text{ lines})$$

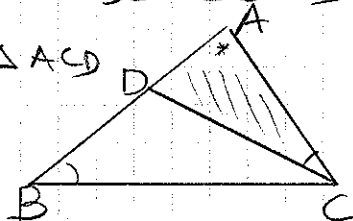
$$107^\circ + 129^\circ + 3x + 10^\circ = 360^\circ \quad (\angle\text{'s at a pt})$$

$$\therefore 3x = 114$$

$$x = 38^\circ$$

Prove

b) $\angle ABC \parallel \angle ACD$



In $\triangle ABC$ and $\triangle ACD$

$$\angle ABC = \angle ACD \quad (\text{given})$$

$$\angle BAC = \angle CAD \quad (\text{common})$$

$$\therefore \angle ACB = \angle ADC \quad (\text{sum of } \angle\text{'s in } \triangle)$$

$$\therefore \triangle ABC \parallel \triangle ACD$$

$$\text{ii) } \triangle ABC \parallel \triangle ACD$$

$$\frac{AB}{AC} = \frac{AC}{AD} = \frac{BC}{CD} \quad \checkmark$$

$$AB \cdot AD = AC^2 \quad \text{as required (1)}$$

$$\text{iii) } AC^2 = AB \times AD$$

$$4^2 = AB \times 4$$

$$AB = 4 \text{ cm} \quad \checkmark \quad (1)$$

$$\text{c) } 2x - 3 \geq 0 \quad \checkmark$$

$$2x \geq 3$$

$$x \geq \frac{3}{2} \quad \checkmark \quad (2)$$

$$\text{d) } f(x) = \begin{cases} -1 & x \leq -2 \\ 2x & -2 < x < 0 \\ 3 - x^2 & x \geq 0 \end{cases}$$

$$f(-2) + f\left(\frac{1}{2}\right) + 2f(4)$$

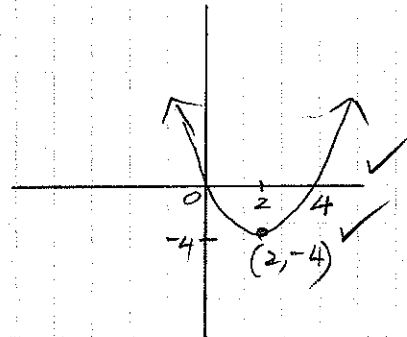
$$= (-1) + \left[3 - \left(\frac{1}{2}\right)^2\right] + 2\left[3 - (4)^2\right]$$

$$= -1 + \frac{11}{4} + 2(-13)$$

$$= -1 + \frac{11}{4} - 26$$

$$= -24\frac{1}{4} \quad \checkmark \quad (3)$$

e)



$$y = x^2 - 4x$$

$$y = x(x - 4)$$

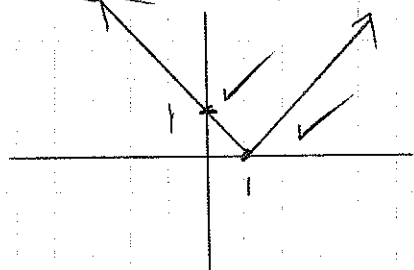
$$\text{Subst } x = 2$$

$$y = (2)^2 - 4(2)$$

$$= -4 \quad (2)$$

ii)

$$y = |x - 1|$$



(2)

TOTAL = 15

Question 15:

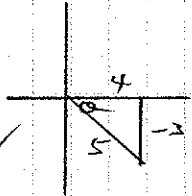
$$\begin{aligned} \text{a) } \cos 135^\circ &= -\cos 45^\circ \\ &= -\frac{1}{\sqrt{2}} \quad \checkmark \end{aligned}$$

(1)

$$\begin{aligned} \text{ii) } \tan(-60^\circ) &= -\tan 60^\circ \\ &= -\sqrt{3} \quad \checkmark \end{aligned}$$

(1)

$$\begin{aligned} \text{b) } \tan \theta &= -\frac{3}{4} \\ \cos \theta &= \frac{4}{5} \quad \checkmark \end{aligned}$$



(2)

$$\begin{aligned} \text{c) } 2\sin \theta + 1 &= 0 \\ \sin \theta &= -\frac{1}{2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \theta &= 180^\circ + 30^\circ & \theta &= 360^\circ - 30^\circ \\ &= 210^\circ & &= 330^\circ \quad \checkmark \end{aligned}$$

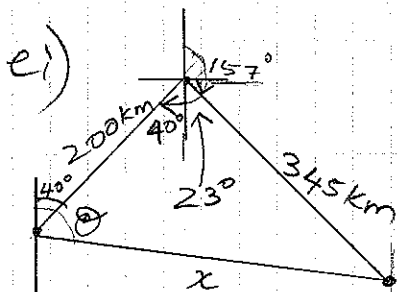
(2)

$$\text{d) } \frac{\sin(90^\circ - \theta)}{\sin(180^\circ - \theta)}$$

$$= \frac{\cos \theta}{1 \cdot \sin \theta} \quad \checkmark$$

$$= \cot \theta \quad \checkmark$$

(2)



(1)

$$\text{ii) } x^2 = 200^2 + 345^2 - 2 \times 200 \times 345 \times \cos 63^\circ$$

$$x^2 = 96374.3$$

$$x = 310 \text{ km} \quad \checkmark$$

(2)

$$\text{iii) } \frac{\sin \theta}{345} = \frac{\sin 63^\circ}{310} \quad \checkmark$$

$$\sin \theta = \frac{345 \times \sin 63^\circ}{310} \quad (2)$$

$$\theta = 83^\circ \quad \checkmark$$

$$\text{f) } \frac{a}{\sin 30^\circ} = \frac{10}{\sin 45^\circ}$$

$$a = \frac{10 \times \sin 30^\circ}{\sin 45^\circ} \quad \checkmark$$

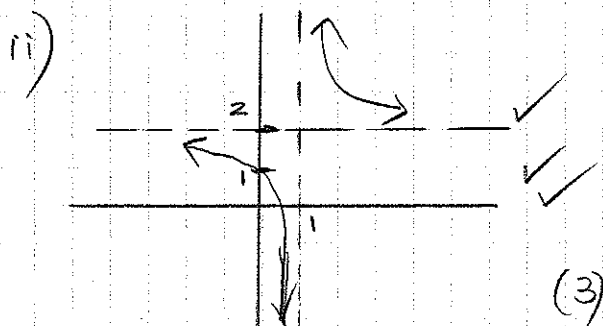
$$= \frac{10 \times \frac{1}{2}}{\frac{1}{\sqrt{2}}}$$

$$a = \underline{\underline{5\sqrt{2}}} \quad \checkmark \quad (2)$$

TOTAL = 15

Question 16:

$$\begin{aligned}
 \text{a) } f(-x) &= \frac{1}{-x-1} + 2 \\
 &= \frac{1}{-(x+1)} + 2 \\
 &\neq f(x) \quad \checkmark \\
 &\neq -f(x) \quad \checkmark \\
 \therefore \text{neither} \quad \checkmark \quad (3)
 \end{aligned}$$

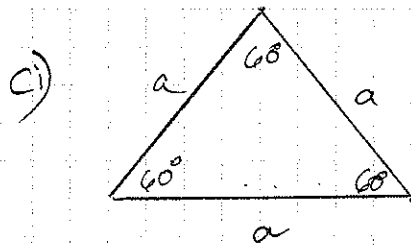


iii) Domain: All real numbers $x \neq -1$ ✓

Range: All real numbers $y \neq 2$ ✓ (2)

iv) Vertical Asymptote: $x = -1$ ✓ (1)

$$\begin{aligned}
 \text{b) } \underline{\text{LHS}} &: \tan \theta = \frac{\sin \theta}{\cos \theta} \\
 &= \frac{\sin \theta}{\cos \theta} - \frac{\sin^3 \theta}{\cos^3 \theta} \quad \checkmark \\
 &= \frac{\sin \theta (1 - \sin^2 \theta)}{\cos^3 \theta} \quad \checkmark \\
 &= \frac{\sin \theta (\cos^2 \theta)}{\cos^3 \theta} \quad \checkmark \\
 &= \sin \theta \cdot \cos \theta \\
 &= \text{RHS} \quad (2)
 \end{aligned}$$



$$\begin{aligned}
 \text{Area} &= \frac{1}{2} ab \sin C \\
 &= \frac{1}{2} a \cdot a \cdot \sin 60^\circ \quad \checkmark \\
 &= \frac{1}{2} a^2 \frac{\sqrt{3}}{2} \quad \checkmark \\
 &= \frac{a^2 \sqrt{3}}{4} \quad (2)
 \end{aligned}$$

$$\text{ii) Area} = 16\sqrt{3}$$

$$\therefore \frac{a^2 \sqrt{3}}{4} = 16\sqrt{3} \quad \checkmark$$

$$\therefore \frac{a^2}{4} = 16$$

$$a^2 = 64$$

$$\underline{\underline{a = 8}} \quad \checkmark$$

$$\therefore \underline{\text{length of its side} = 8}$$

$$\text{TOTAL} = 15 \quad (2)$$